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Portfolio: romia.dev | case studies, architecture breakdowns, and live demos

Summary

AI Engineer building production multi-agent and Retrieval-Augmented Generation (RAG) systems with full-stack breadth: FastAPI and Spring Boot backends, Kafka event-driven microservices, and React/Next.js. Co-founded Zaylon AI and built its LangGraph tool-calling agent and tri-lingual NLP on a multi-tenant FastAPI backend.

Experience

Software Engineer & Co-Founder

May 2025 – Present

Zaylon AI

- Lead engineering (architecture, reviews, roadmap, DevOps); built the **FastAPI** backend with **Docker** and CI/CD.
- Built the conversational AI as a single **LangGraph** tool-calling agent with around 30 scoped tools.
- Engineered **DialectBridge**: tri-lingual NLP with dialect detection for English, Egyptian Arabic, and Franco-Arabic.
- Hardened the backend: **argon2id** auth, row-level multi-tenant isolation over 80+ tables, a Postgres checkout saga.

Software Integration Engineering Intern

Jul 2025

Ejada

- Built **Spring Boot** REST APIs for core banking with query optimization and **Redis** caching.
- Ran **Kafka** producers/consumers for async messaging; secured the gateway with OAuth2 and rate limiting (WSO2).

AI Coding Expert

Sep 2024 – Feb 2026

Freelance

- Ranked thousands of Python samples and built quality rubrics for RLHF pipelines training frontier AI models.

Projects

Case studies, architecture breakdowns, and live demos at romia.dev

Talos: Multi-User RAG Document Platform

Graduation Project (Team)

- Owned the ingestion and retrieval core of a **FastAPI** document-chat platform; cited answers streamed over SSE.
- Hybrid retrieval (dense + **BM25**, RRF) with cross-encoder rerank; async **ARQ/MinIO/Milvus** ingestion, eval harness.

Virtual Banking Microservices

github.com/Ab-Romia/Virtual-Bank-System

- Five **Spring Boot 3** services behind a gateway; **Kafka** transfer saga: outbox, idempotent consumers, RS256 JWT.
- A Testcontainers test fires 20 concurrent transfers and proves no double-spend, never a negative balance.

ContextIQ: Hybrid-Retrieval RAG

github.com/Ab-Romia/ContextIQ-RAG

- **FastAPI** RAG on CPU (ONNX): dense + **BM25** fused with RRF, cross-encoder reranking, grounded citations.
- Measured on a confusable corpus: hit@3 0.83 and MRR 0.78 against a dense-only 0.67 baseline.

Multimodal Emotion Recognition

github.com/Ab-Romia/RAVDESS-emotion-recognition

- Speaker-independent (actor-disjoint): frozen **WavLM-large** + facial-expression model, calibrated late fusion.
- 70.3% audio-only, 78.8% audio-visual on 8 emotions, leak-free (random splits inflate audio to around 78%).

VoicePrint: Authorship Stylometry

huggingface.co/spaces/Ab-Romia/voiceprint

StyleDistance embedding plus a function-word vector; 0.889 accuracy across five authors vs a 0.20 baseline.

Education

Alexandria University, Faculty of Engineering

Expected Jul 2026

B.Sc. in Computer & Communications Engineering (AI Concentration) | Cumulative GPA: 3.73 / 4.0

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, C/C++

Backend: FastAPI, Spring Boot, Microservices, Event-Driven Architecture, Kafka, REST APIs, PostgreSQL, Redis

AI / ML: LangGraph, LangChain, PyTorch, RAG, Multi-Agent Systems, NLP, Vector DBs (ChromaDB, pgvector), Reranking

Frontend & DevOps: React, Next.js, Tailwind CSS, Docker, Git, GitHub Actions, CI/CD, Linux

Certifications: micro1 Certified Software Engineer | [Google AI Agents](https://www.google.com/ai/agents/) | Meta Back-end | AWS Cloud & ML

Competitions: MTC-AIC3 BCI (Egypt National AI) | ECPC (3x) | Hepsiburada NLP Hackathon (TEKNOFEST)

Languages: English (Fluent) | Arabic (Native) | German (A2) | Spanish (Conversational)